

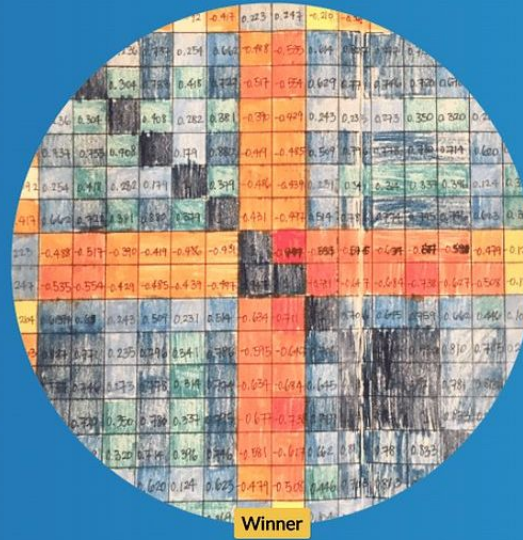
Today is the infrastructure day, not a real lecture.

By the end: you'll know what the words mean, your tools will work, and you'll have a group.

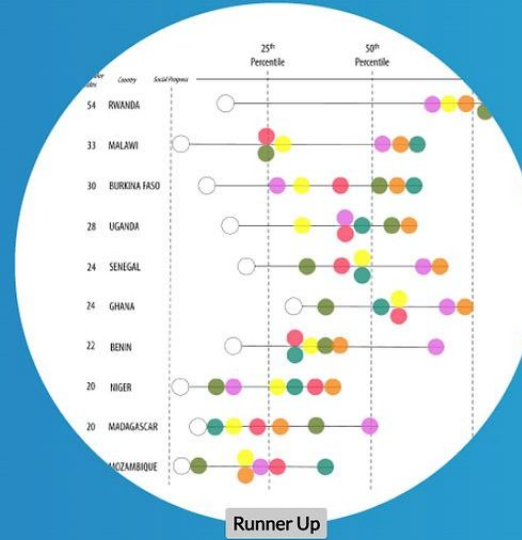




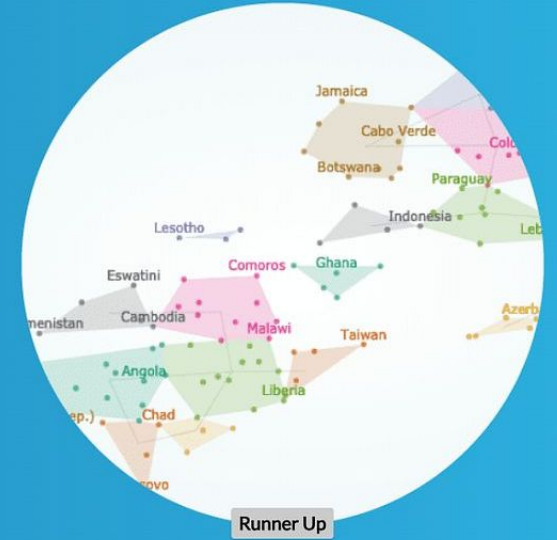
GOV | DNA
By Werner Helmich



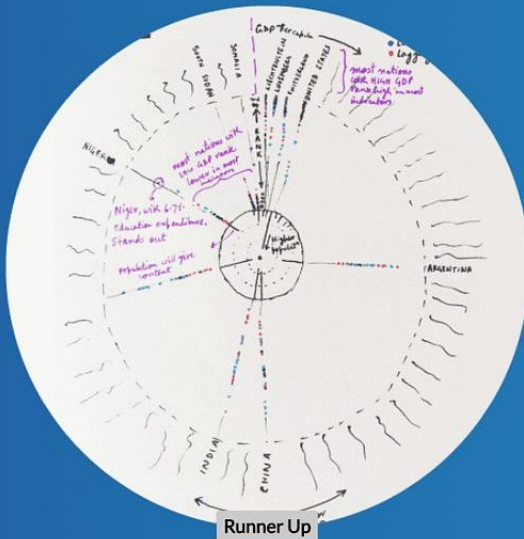
What Makes The Government Effective?
by Eileen Huang



The Value Proposition of Good Government
By Anna Jacobson



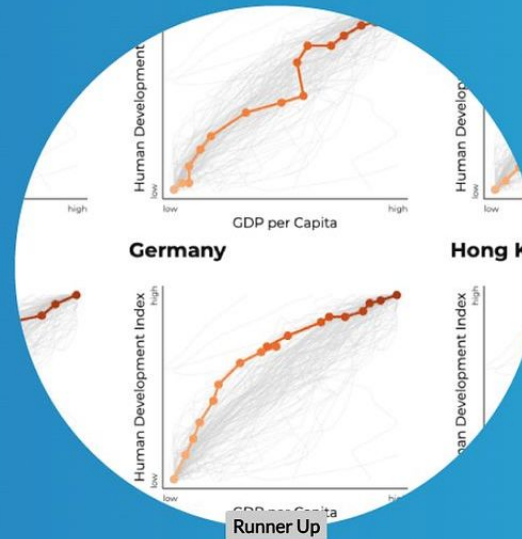
Mapamundi
By Christian Parsons



Worst Of The Best, Best Of The Worst
By Rasagy Sharma



Smaller? Better? More Productive?
By Dimitar Toshkov



Good Governments Help People Succeed
By Jim Vallandingham



An Alternative Data-Driven World Map
By Interacta

Source: <https://informationisbeautiful.net/>



Who are you?

- Your name
- Your interest (inside or outside computing)
- Have you decided on your final BSc project?
- What do you want to get out of this course?

Overview of this course

Week	Date	Topic
1	07.09.2026	Introduction
2	14.09.2026	Data, Research Questions & Ethics
3	21.09.2026	Data Visualisation & Exploratory Data Analysis
4	28.09.2026	Preprocessing & Feature Engineering
5	05.10.2026	Machine Learning Fundamentals & Regression
6	12.10.2026	Project Presentations I
7	19.10.2026	Classification: Logistic Regression, KNN & Decision Trees
8	26.10.2026	Bank Holiday — No Class
9	02.11.2026	Gradient Descent, Regularisation & Model Complexity
10	09.11.2026	Model Evaluation & Ensemble Methods
11	16.11.2026	Neural Networks, Unsupervised Learning & Special Topics
12	23.11.2026	Project Presentations II

Rules and Regulations

- Why are we here?
- Use of AI
- Taking responsibility



Setting up the infrastructure

We need this working **today**:

- 1 — Your own laptop, or a lab machine you'll use consistently all term
- 2 — A **GitHub** account: share your handle with us on the designated Loop link.
- 3 — **GitHub Desktop** installed
- 4 — **VS Code** installed

 **Why:** an analysis you can't re-run is an analysis nobody can check, including you, in three weeks.

Why git? Why version control?

thesis_final.pdf

thesis_final_final.pdf

thesis_final_final_final_theLastIPromise.pdf

*thesis_f*_final.pdf*

Git was written by Linus Torvalds in about two weeks in 2005. Its own manual page calls it the stupid content tracker, and "git" is British slang for an unpleasant person.

Peng (2011), Science 334:1226–1227 · Ram (2013), Source Code Biol Med 8:7 · Bryan (2018), The American Statistician 72(1):20–27

GitHub/GitLab vs git

git is the version control tool that runs locally; **GitHub** and **GitLab** are cloud platforms that host git repositories and add collaboration features.



GitHub



GitLab



A typical weekly Git Repo

NoCh-Git Initial commit 3f98a27 · 3 minutes ago 1 Commit		
data	Initial commit	3 minutes ago
notebooks	Initial commit	3 minutes ago
slides	Initial commit	3 minutes ago
.gitignore	Initial commit	3 minutes ago
LICENSE	Initial commit	3 minutes ago
README.md	Initial commit	3 minutes ago
requirements.txt	Initial commit	3 minutes ago



**IN CASE
OF FIRE**

> git commit
> git push
> git out of here

locaweb

What is data?

- What **is** data?
- Where do we **find** it?
- What kind of data do **you** work with daily — and **who** **decided what got recorded?**

Data is the plural of the Latin *datum* — "a thing given". The word once meant what you grant at the start of an argument; now it means what you go out and collect.

Data is never simply found. It's always the product of somebody's decision about what to record, at what resolution, and what to leave out.

Rosenberg (2013), in Gitelman (ed.), *"Raw Data Is an Oxymoron"*, MIT Press

Where to find data for your project

A few suggestions:

National statistics office ·
government open-data
portal

Eurostat

OpenML

Kaggle

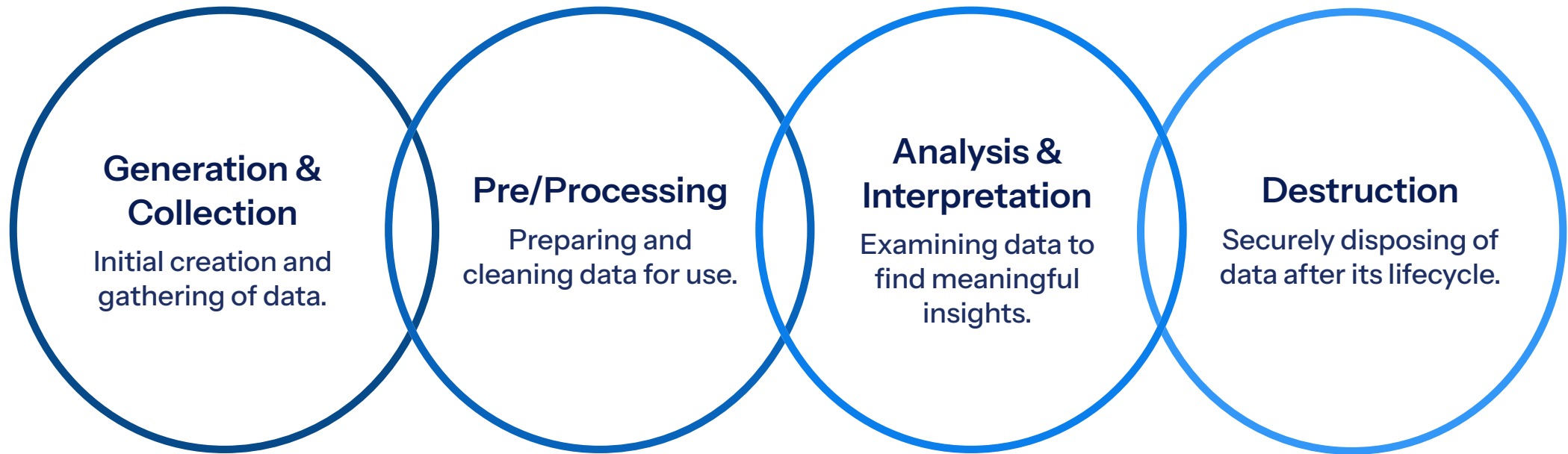
Zenodo

Our World in Data



Requirement: it must come with documentation.

The data lifecycle



- **Not a straight line.** You will go backwards, repeatedly.
- **Destruction can be a legal obligation, not an afterthought.** Under GDPR, personal data should not be kept longer than necessary.

Adapted from Wing (2019), Harvard Data Science Review 1(1) plus GDPR Arts. 5(1)(e), 17

Types of data

Structured

Rows and columns, fixed schema,
fits a database

*Transactions, sensor logs, survey
responses*

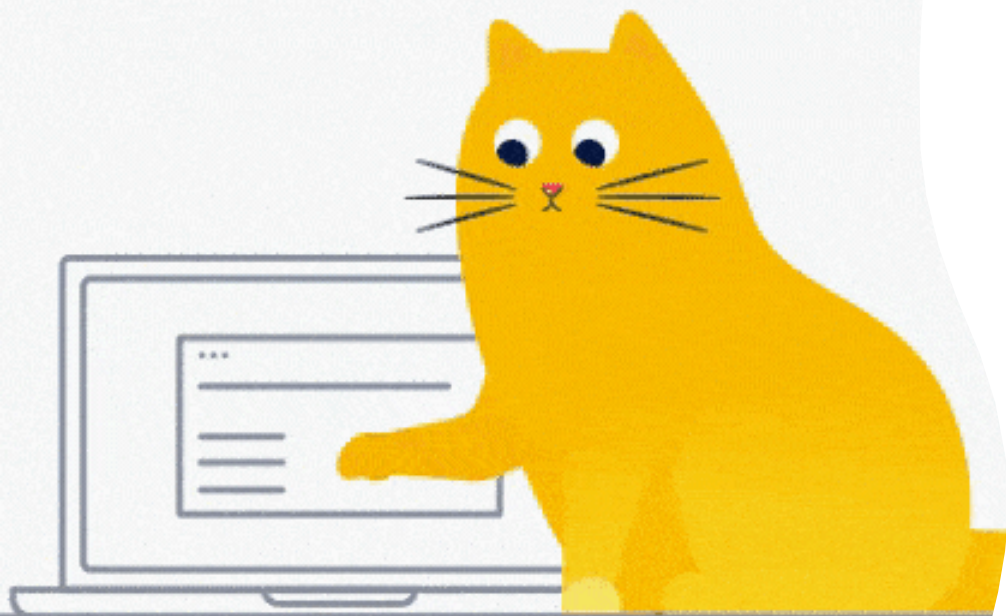
Semi-structured

Has structure, but no fixed schema

JSON, XML, log files...

Unstructured

Text, images, audio, video, free-
text notes



Who here has programming experience?

- Which **languages**?
- Have you used **Git** before?
- Have you used **pandas**, **R**, or **Excel** for analysis?

 **This course uses Python. You do not need to be good at it yet.**

Who has ever done an analysis entirely by clicking around in a spreadsheet, and could reproduce it exactly today?

Classical programming vs machine learning

Classical

rules + data $\rightarrow$ answers

Machine learning

data + answers $\rightarrow$ rules

Use ML when the rules are **unknown**, **too numerous**, or **keep changing**.

If you can write the rule, write the rule. A four-line if statement never needs a training set, a GPU, or a fairness audit.

Many ML problems share one property: **the mapping is difficult to specify explicitly**. Nobody can specify in closed form what makes a photo contain a cat.

Ask the room: one task from your part-time job that should *not* be ML, and one that should.

Domingos (2012), CACM 55(10):78–87

Four words used interchangeably (they shouldn't)

Statistics

Quantifying variation and uncertainty; making inferences from data

What's going on, and how certain are we?

Machine learning

Algorithms that improve at a task with data

Will this predict well?

Data mining

Discovering patterns in large data

What's in here that we didn't know?

Data analytics

Turning data into decisions

So what do we do?

statistics is often concerned with **inference and explanation**, ML with **prediction**, data mining with **pattern discovery**, and analytics with **using data to answer practical questions**. The areas have considerable overlap but are not identical.

Course framing, informed by Breiman (2001), Statistical Science 16(3):199–231 · Donoho (2017), JCGS 26(4):745–766

The toy project

- 1 Groups of 3 (maximum 4)
- 2 A real dataset and real research questions you care about. This runs across the whole term.
- 3 Form your groups this week.
- 4 Nominate someone responsible for the repository.

What kind of data are we aiming to analyze in this course?



Structured, tabular

1

Large enough for meaningful analysis and modeling, preferably > ~10,000 records



Documented: you should be able to say where every column came from



Legally and ethically usable: no scraped personal data

How to ask a good research question

A good research question passes the FINER criteria (Hulley et al., 2013):

-  **Feasible:** with your data, your skills, and one term
-  **Interesting:** to someone other than you
-  **Novel:** adds something beyond what is already known
-  **Ethical:** you'd be comfortable if the subjects read your report
-  **Relevant:** the answer changes something

Bad: "What can we learn about gardening?"

Better: "Do violets planted in winter produce fewer blossoms in their first year than violets planted in summer?"

The test: a good question tells you what data would answer it, and what answer would prove you wrong.

Until next week

01

Make sure you can run this week's repo. If it doesn't work, tell us before Friday and find a solution BEFORE next session.

02

Form your groups (ideally 3, max 4) and enter the names in Loop

03

Draft your research questions: 2, maximum 3.



References and read mores

- Breiman, L. (2001). Statistical modeling: the two cultures. *Statistical Science*, 16(3), 199–231.
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